Tribhuvan University

Faculty of Humanities and Social Science

A PROJECT PROPOSAL REPORT ON

EduTender - Tender Management System

Submitted to:

Department of Computer Application

Mechi Multiple Campus

In partial fulfillment of the requirements for the Bachelor in Computer Application

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Table of Contents

List of Figures

1. Introduction

The Edu-Tender System is a website that helps schools and colleges buy things they need. This system lets them put up notices on the website and companies can send in their bids over the internet. The E-Tender System has a place where people can make tenders send in bids look at bids and pick the company they want to work with. This makes buying things more open and safe for everyone.

The proposed system aims to develop an E-Tender Management System for universities and colleges. This means schools can make and look after tenders on the website and companies can look at tenders and send in bids to try to win the job. The E-Tender Management System has lots of features like putting tenders online, secret bidding looking at bids and picking a winner. All of this helps make sure the buying process is fair and open.

The E-Tender System is good because it reduces the amount of paper work makes things easier for people to do and helps everyone get to the website. This makes the buying process better and more reliable for schools and companies. The E-Tender System is a tool, for educational institutions and vendors to use when they need to buy or sell things.

2.Problem Statement

Educational institutions still have a lot of trouble with procurement and tendering. They use methods that do not work well. Here are some of the issues:

- Traditional paper-based tender management is time-consuming and inefficient.
- Lack of a centralized platform for tender publication and bid submission reduces efficiency. This makes procurement slow.
- Keeping track of tender records is hard. This makes it difficult to look back at tenders.
- Checking and comparing bids by hand can lead to mistakes and delays.

To fix these problems we have come up with a system called the "E-Tender Management System"

3.Objectives

The main goals of the E-Tender Management System are listed below:

- To automate the tender creation and bidding process
- To reduce the paperwork and processing time
- To improve transparency, efficiency, and procurement record management.

4.Methodology

4.1The Iterative Waterfall Model

This project is developed using Iterative Waterfall Methodology. Planning, Analysis, Design, Implementation, Testing, and Maintenance are performed iterative and Iterative Waterfall does not adapt dynamically like Agile. Thus, this model is efficient.

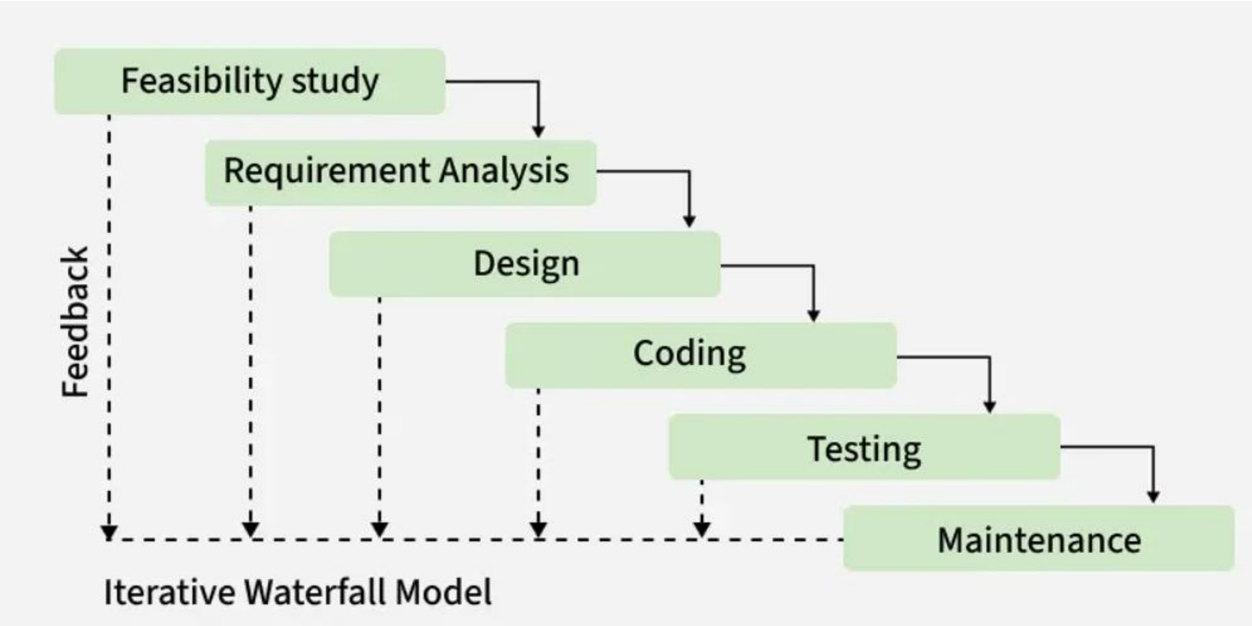


Figure 4.1 : Iterative Waterfall Methodology for Edu-Tender

Iterative Waterfall Model is a software development approach that combines the sequential steps of the traditional Waterfall Model with the flexibility of iterative design. It allows for improvements and changes to be made at each stage of the development process, instead of waiting until the end of the project. The iterative waterfall model provides feedback paths from every phase to its preceding phases, which is the main difference from the classical waterfall model [1]

4.2 Requirement

Under the topic following requirement are identified and research for the development of project:

4.2.1 Study of existing system

On the study of "EduTender" a E-Tender System, the major issues identified are listed as below:

- Slow and time consuming manual procedure.
- Lack of transparency and fairness.

4.2.2 Functional Requirements

The functional requirements that are included in the system are shown in the following figure:

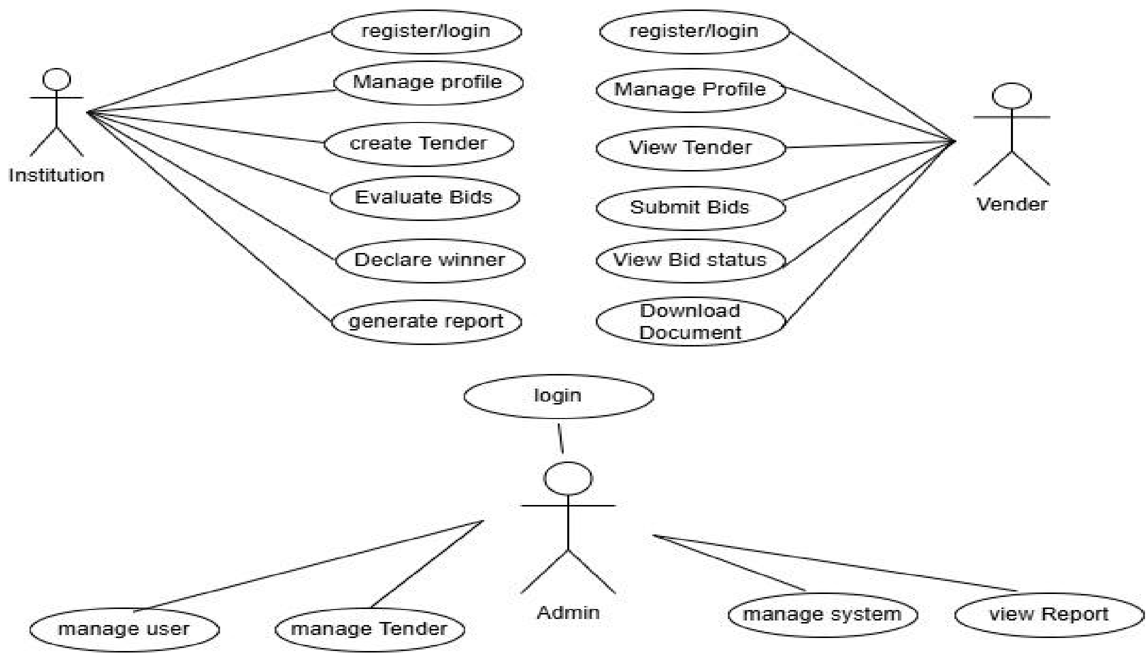


Figure4.2.2: Use-Case Diagram

The Institution (School/University) Perspective

- **Register/Login:** Create an account and securely access the administrative dashboard.
- **Manage Profile:** Update institutional information, contact details, or credentials.
- **Create Tender:** Draft and publish a request with item specifications, budget limits, and deadlines.
- **Evaluate Bids:** Review vendor applications, pricing details, and qualification documents after deadlines.
- **Declare Winner:** Award the contract to the selected vendor and automatically notify all participants.
- **Generate Reports:** Export summary files of bidding history and final decisions for record-keeping.

The Company (Vendor) Perspective

- **Register/Login:** Create a business profile and securely log into the platform.
- **Manage Profile:** Update corporate documentation, business descriptions, or contact details.
- **View Tender Details:** Review requirements, rules, and deadlines for specific active tenders.
- **Submit Bid:** Input a price offer, upload corporate documents, and submit the proposal online.
- **View Bid Status:** Track whether a submitted proposal is pending, accepted, or rejected.
- **Download Documents:** Access guidelines or official contract templates attached to the tender.

The Admin Perspective

- **Login:** Sign into a master dashboard with elevated privileges to oversee the system.
- **Manage Users:** Approve accounts, block fake users, and handle password resets for schools or vendors.
- **Manage Tenders:** Monitor active notices for appropriate content; edit or remove them upon request.
- **Manage System:** Maintain website health, update security settings, and ensure database stability.
- **View Reports:** Access platform analytic (e.g., total bids, monthly active users) to evaluate performance.

4.2.3 Non-Functional Requirements

Non-functional requirements (NFRs) define system attributes such as security, reliability, performance, maintainability, scalability, and usability [2]. The non-functional requirements of the E-Tender Management System are:

- **Reliability:** The system shall ensure stable performance and continuous availability during tender publication and bid submission.

- **Maintainability:** Being entirely web-based, the platform requires no specialized physical hardware. The system architecture allows easy maintenance, updates, and bug fixes.
- **Scalability:** The platform efficiently handles data fluctuations during heavy, deadline-driven bidding periods. Its optimized online database minimizes processing loads, allowing easy scaling to accommodate more colleges and vendors over time.
- **Usability:** The system features an intuitive, user-friendly interface with a clear, guided look for step-by-step bidding operations. No professional certifications or advanced engineering skills are required for school staff or vendors to navigate it.

4.3 Feasibility Study

Under feasibility study the system will primarily focus on technical feasibility, operational feasibility and economic feasibility which are explained as below:

4.3.1 Technical Feasibility

This system will be built using existing technologies. Visual Studio Code will be used as the primary coding software. HTML, CSS, and JavaScript will be used for the Front-End development of the system to ensure an intuitive interface. While for Back-End language and database management, Django and SQLite3 will be used. For the development, a laptop, mouse, and keyboard will be used to complete the project.

4.3.2 Operational Feasibility

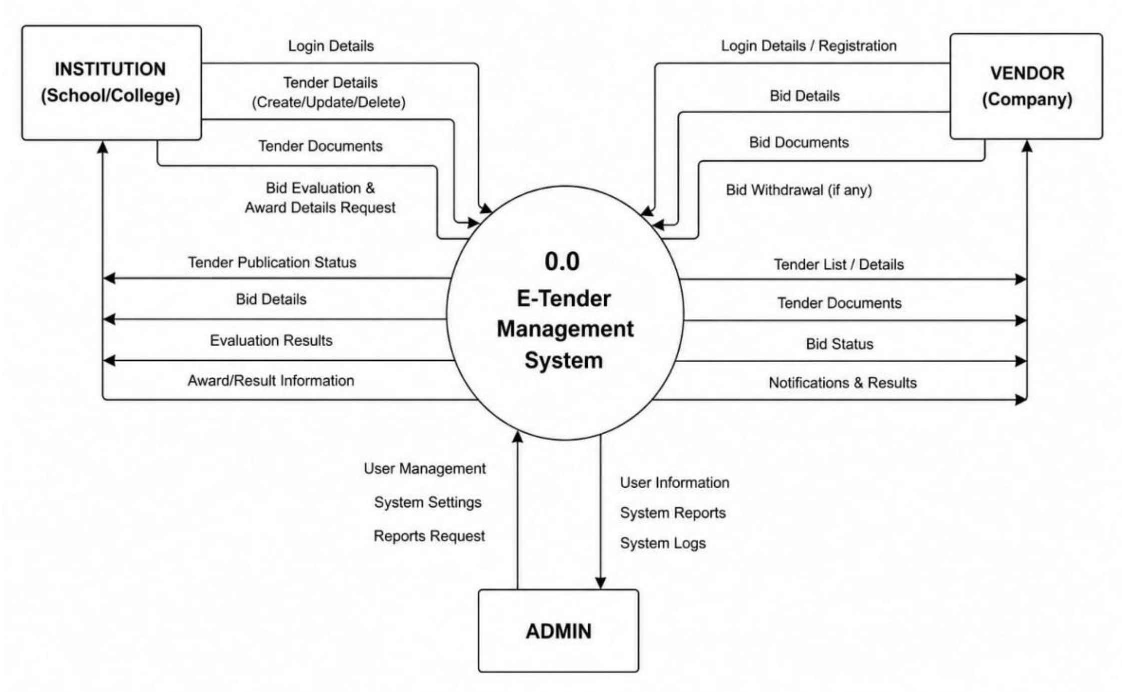
The system will be secure, user-friendly, and fully responsive on both mobile phones and laptops. Vendors, contractors, and internal procurement officers who have a basic understanding of computers and internet browsing can easily operate the system, submit bids, and track tender statuses without requiring extensive technical training.

4.3.3 Economic Feasibility

Since it is a college-level project, it doesn't require any extra expenses other than internet access and a device to develop and access the website. Furthermore, using this software for organizations will be economically beneficial since it eliminates massive paperwork, reduces administrative labour costs, and reduces processing time previously used to manually distribute, collect, and evaluate physical tender documents.

4.4 High Level Design of System

4.4.1 System Architecture (Context DFD)



Figures 4.4.1: DFD Diagram of Edu-Tender

The diagram above models the high-level boundary of the EduTender, illustrating data exchanges between institutions, company, admin and the system core.

4.4.2 ER Diagram

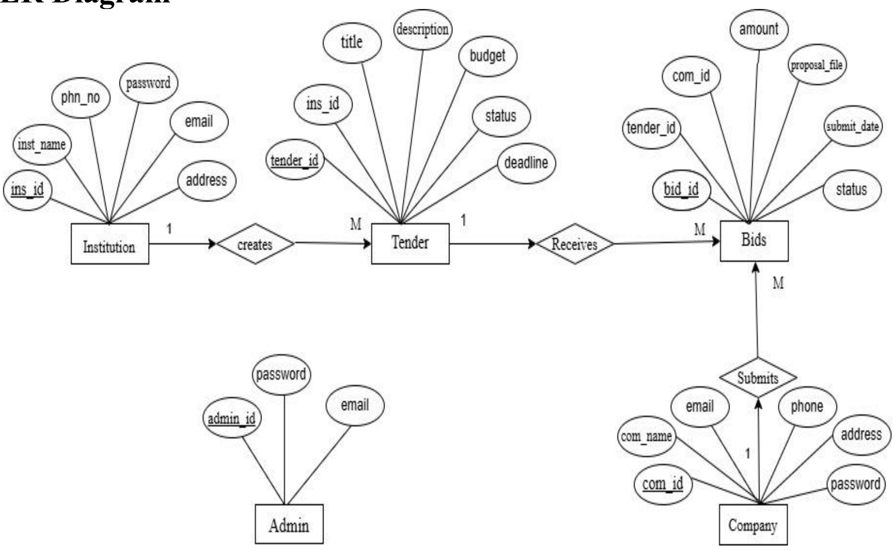


Figure 4.4.2:Entity-Relation Diagram

Figure 4.3.1 represents the ER (Entity Relation) Diagram model of the project system. The rectangle shaped box are entities, oval shape represents attributes and diamond shape represents relationship between entities.

5. Gantt Chart

The tentative time schedule to accomplish this project along with task are as shown in below figure:

Figure 5: Gantt Chart

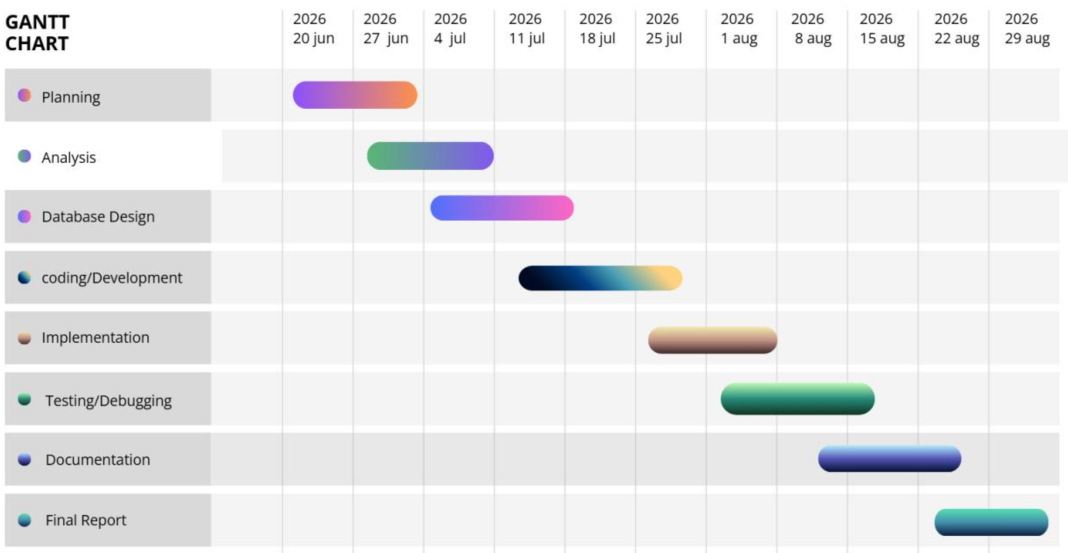


Figure 5 represents the timeline of the project. The phases will be completed on average of specified days. Some phase will take time and it may take several other days. It includes planning, analysis, designing, coding, testing and implementation. These phases will be carried out on following dates shown on the figure. Since Agile Methodology is used for the project, the process can be repeated on these days

6. Expected Outcomes

The end goal of the E-Tender Management System is to have a procurement process that is straightforward, open, and completely digital. This means schools can count on getting the best vendor for the job at a fair price, thanks to a thorough and secure evaluation process. On the other hand, companies get to compete on an equal footing, easily finding and applying for opportunities, and tracking how their bids are doing. In the end, the system gives administrators a tidy, mistake-free way to handle transactions, keep accurate records, and protect everyone's data. This way, everything runs smoothly and securely for all parties involved

7. References

[1] GeeksforGeeks, "Iterative waterfall model" [online] Available: https://www.geeksforgeeks.org/software-engineering-iterative-waterfall-model/ [Accessed 20 June 2026]

[2] Scaned Agile, "Scaled Agile Framework," [Online]. Available: https://www.scaledagileframework.com/nonfunctional-requirements/ [Accessed 20 June 2026].